

Important Information. Please read carefully.



Composition

Each capsule contains Omeprazole 20mg in the form of enteric-coated pellets.

Pharmacodynamics

Omeprazole is a proton pump inhibitor. It suppresses secretion of gastric acid by inhibiting the enzyme system of hydrogen/ potassium adenosine triphosphatase (H⁺/K⁺ ATPase), the ‘proton pump’ of the gastric parietal cell.
During treatment with antisecretory medicinal products, serum gastrin increases in response to the decreased acid secretion. Also, Chromogranin A (CgA) increases due to decreased gastric acidity. The increased CgA level may interfere with investigations for neuroendocrine tumours.
Available published evidence suggests that proton pump inhibitors should be discontinued between 5 days and 2 weeks prior to CgA measurements. This is to allow CgA levels that might be spuriously elevated following PPI treatment to return to reference range.

Pharmacokinetics

Absorption

Omeprazole and omeprazole magnesium are acid labile and are therefore administered orally as enteric-coated granules in capsules or tablets. Absorption of omeprazole is rapid, with peak plasma levels occurring approximately 1-2 hours after dose. Absorption of omeprazole takes place in the small intestine and is usually completed within 3-6 hours.
Concomitant intake of food has no influence on the bioavailability. The systemic availability (bioavailability) from a single oral dose of omeprazole is approximately 40%. After repeated once-daily administration, the bioavailability increases to about 60%.

Distribution

The apparent volume of distribution is approximately 0.3 l/kg body weight. Omeprazole is 97% plasma protein bound.

Metabolism

Omeprazole is completely metabolised by the cytochrome P450 system (CYP). The major part of its metabolism is dependent on the polymorphically expressed CYP2C19, responsible for the formation of hydroxyomeprazole, the major metabolite in plasma. The remaining part is dependent on another specific isoform, CYP3A4, responsible for the formation of omeprazole sulfone. As a consequence of high affinity of omeprazole to CYP2C19, there is a potential for competitive inhibition and metabolic drug-drug interactions with other substrates for CYP2C19. However, due to low affinity to CYP3A4, omeprazole has no potential to inhibit the metabolism of other CYP3A4 substrates. In addition, omeprazole lacks an inhibitory effect on the main CYP enzymes. Approximately 3% of the Caucasian population and 15-20% of Asian populations lack a functional CYP2C19 enzyme and are called poor metabolisers. In such individuals the metabolism of omeprazole is probably mainly catalysed by CYP3A4. After repeated once-daily administration of 20 mg omeprazole, the mean AUC was 5 to 10 times higher in poor metabolisers than in subjects having a functional CYP2C19 enzyme (extensive metabolisers). Mean peak plasma concentrations were also higher, by 3 to 5 times.

Elimination

The plasma elimination half-life of omeprazole is usually shorter than one hour both after single and repeated oral once daily dosing. Omeprazole is completely eliminated from plasma between doses with no tendency for accumulation during once-daily administration. Almost 80% of an oral dose of omeprazole is excreted as metabolites in the urine, the remainder in the faeces, primarily originating from bile secretion.

Indication

Zenpro Capsule is indicated for:

- Treatment of duodenal ulcer
- Treatment of gastric ulcer
- Treatment of NSAID associated gastric and duodenal ulcers or erosions
- Treatment of Helicobacter pylori eradication in peptic ulcer disease
- Treatment of Reflux oesophagitis
- Treatment of Symptomatic gastro-oesophageal reflux disease
- Treatment of Zollinger-Ellison syndrome
- Treatment of Acid related dyspepsia

Dosage

Duodenal ulcer

The recommended dosage in patients with an active duodenal ulcer is omeprazole 20mg once daily. Symptom resolution is rapid and in most patients healing occurs within 2 weeks. For those patients who may not be fully healed after the initial course, healing usually occurs during a further 2 week treatment period. In patients with poorly responsive duodenal ulcer omeprazole 40mg once daily is recommended and healing is usually achieved within 4 weeks. For the prevention of relapse in patients with duodenal ulcer disease the recommended dose is omeprazole 10 mg once daily. If needed the dose can be increased to omeprazole 20-40mg once daily. For NSAID associated duodenal ulcers see NSAID associated gastroduodenal lesions. For eradication of Helicobacter pylori see Helicobacter pylori (Hp) eradication regimens in peptic ulcer disease.

Gastric ulcer

The recommended dosage is omeprazole 20 mg once daily. Symptom resolution is rapid and in most patients healing occurs within 4 weeks. For those patients who may not be fully healed after the initial course, healing usually occurs during a further 4 weeks treatment period. In patients with poorly responsive gastric ulcer, omeprazole 40 mg once daily is recommended and healing is usually achieved within 8weeks. For the prevention of relapse in patients with poorly responsive gastric ulcer the recommended dose is omeprazole 20 mg once daily. If needed the dose can be increased to omeprazole 40mg once daily. For eradication of Helicobacter pylori see Helicobacter pylori (Hp) eradication regimens in peptic ulcer disease.

NSAID associated ulcers or gastroduodenal erosions

NSAID associated gastric ulcers, duodenal ulcers or gastroduodenal erosions in patients without continued NSAID treatment the recommended dosage of omeprazole is 20mg once daily. Symptom resolution is rapid and in most patients healing occurs within 4 weeks. For those patients who may not be fully healed after the initial course, healing usually occurs during a further 4 weeks treatment period. For the prevention of NSAID associated gastric ulcers, duodenal ulcers, gastroduodenal erosions and dyspeptic symptoms the recommended dosage of omeprazole is 20mg once daily.

Helicobacter pylori (Hp) eradication regimens in peptic ulcer disease

Triple therapy regimens:

Omeprazole 20mg, amoxicillin 1g and clarithromycin 500mg, all twice a day for one week or
Omeprazole 20mg, metronidazole 400mg (or tinidazole 500mg) and clarithromycin 250mg, all twice a day for one week, or
Omeprazole 40mg once daily with amoxicillin 500mg and metronidazole 400mg both three times a day for one week.

Dual therapy regimens:

Omeprazole 40-80mg daily with amoxicillin 1.5g daily in divided doses for two weeks. In clinical studies daily doses of 1.5-3g of amoxicillin have been used, or
Omeprazole 40mg once daily and clarithromycin 500mg three times a day for two weeks.

To ensure healing in patients with active peptic ulcer disease, see further dosage recommendations for Duodenal and Gastric ulcer.

In each regimen if the patient is still Hp positive, therapy may be repeated.

Reflux oesophagitis

The recommended dosage is omeprazole 20mg once daily. Symptom resolution is rapid and in most patients healing occurs within 4 weeks. For those patients who may not be fully healed after the initial course, healing usually occurs during a further 4 weeks treatment period.

In patients with severe reflux oesophagitis omeprazole 40mg once daily is recommended and healing is usually achieved within 8weeks.

For the long term management of patients with healed reflux oesophagitis the recommended dose is omeprazole 10mg once daily. If needed the dose can be increased to omeprazole 20-40mg once daily.

Symptomatic gastro-oesophageal reflux disease

The recommended dosage is omeprazole 20mg daily. Symptom relief is rapid. Patients may respond adequately to 10mg daily, and therefore individual dose adjustment should be considered. If symptom control has not been achieved after 4 weeks treatment with omeprazole 20mg daily, further investigation is recommended.

Acid related dyspepsia

In the relief of symptoms in patients with epigastric pain/discomfort with or without heartburn the recommended dosage is omeprazole 20mg once daily.

Patients may respond adequately to 10mg daily and therefore this dose could be considered as a starting dose.

If symptom control has not been achieved after 4 weeks treatment with omeprazole 20mg daily, further investigation is recommended.

Zollinger Ellison Syndrome

In patients with Zollinger - Ellison Syndrome the dosage should be individually adjusted and treatment continued as long as is clinically indicated. The recommended initial dosage is omeprazole 60mg daily. All patients with severe disease and inadequate response to other therapies have been effectively controlled and more than 90% of the patients maintained on doses of omeprazole 20-120mg daily. When doses exceed omeprazole 80mg daily, the dose should be divided and given twice daily.

Impaired renal function

Dose adjustment is not needed in patients with impaired renal function.

Impaired hepatic function

As bioavailability and plasma half-life of omeprazole are increased in patients with impaired hepatic function a daily dose of 10-20mg may be sufficient.

Elderly

Dose adjustment is not needed in the elderly.

Children

Zenpro Capsule is not approved for use in children.

METHOD OF ADMINISTRATION

It is recommended to take Zenpro Capsule in the morning, preferably without food, swallowed whole with half a glass of water. The capsules must not be chewed or crushed.

For patients with swallowing difficulties, patients can open the capsule and swallow the contents with half a glass of water or after mixing the contents in fruit juice, applesauce, or in non-carbonated water. Patients should be advised that the dispersion should be taken immediately (or within 30 minutes) and always be stirred just before drinking and rinsed down with half a glass of water. Alternatively patients can suck the capsule and swallow the pellets with half a glass of water. The enteric-coated pellets must not be chewed."

Contraindications

Zenpro Capsule is contraindicated in patients with known hypersensitivity to any component of the formulation. Omeprazole like other proton pump inhibitors (PPIs) must not be used concomitantly with nelfinavir (see Drug Interactions).

Warnings and Precautions

General

Symptomatic response to therapy with omeprazole does not preclude the presence of gastric malignancy.

Atrophic gastritis has been noted occasionally in gastric corpus biopsies from patients treated long-term with omeprazole.

The possibility of malignancy should be excluded before treatment with omeprazole is instituted.

Use in Pregnancy and Lactation: As with most drugs, omeprazole should be used during pregnancy and lactation only if the potential benefit justifies the potential risk. It is not known whether omeprazole is excreted in human milk.

Pediatric Use: Safety and effectiveness in children have not been established.

Gastrointestinal Infections: Decreased gastric acidity due to any means, including proton pump inhibitors, increases the counts of bacterial normally present in the gastrointestinal tract. Treatment with omeprazole may lead to slightly increased risk of gastrointestinal infections caused by bacteria such as Salmonella, Campylobacter and possibly Clostridium difficile.

Regular Surveillance: Patients on proton pump inhibitor treatment (particularly those treated for long term) should be kept under regular surveillance.

Subacute Cutaneous Lupus Erythematosus (SCL): Proton pump inhibitors are associated with very infrequent cases of subacute cutaneous lupus erythematosus (SCL). If lesions occur, especially in sun-exposed areas of the skin, and if accompanied by arthralgia, the patient should seek medical help promptly and the health care professional should consider stopping Zenpro. SCL after previous treatment with a proton pump inhibitor may increase the risk of SCL with other proton pump inhibitors.

Hypomagnesaemia: Severe hypomagnesaemia has been reported in patients treated with PPI like Zenpro for at least three months, and in most cases for a year. Serious manifestations of hypomagnesaemia such as fatigue, tetany, delirium, convulsions, dizziness and ventricular arrhythmia can occur but they may begin insidiously and be overlooked. In most affected patients, hypomagnesaemia improved after magnesium replacement and discontinuation of the PPI.

For patients expected to be on prolonged treatment or who take PPI with digoxin or drugs that may cause hypomagnesaemia (e.g., diuretics), health care professionals should consider measuring magnesium levels before starting PPI treatment and periodically during treatment.

Fracture: Proton pump inhibitors, especially if used in high doses and over long durations (>1 year), may modestly increase the risk of hip, wrist and spine fracture, predominantly in the elderly or in presence of other recognized risk factors. Observational studies suggest that proton pump inhibitors may increase the overall risk of fracture by 10-40%. Some of this increase may be due to other risk factors. Patients at risk of osteoporosis should receive care according to current clinical guidelines and they should have an adequate intake of vitamin D and calcium.

Clostridium difficile Diarrhea: Published observational studies suggest that PPI therapy may be associated with an increased risk of Clostridium difficile associated diarrhea, especially in hospitalized patients. This diagnosis should be considered for diarrhea that does not improve. Patients should use the lowest dose and shortest duration of PPI therapy appropriate to the condition being treated.

Vitamin B12 Deficiency: Daily treatment with any acid-suppressing medications over a long period of time (e.g. longer than 3 years) may lead to malabsorption of cyanocobalamin (vitamin B12) caused by hypo- or achlorhydria. Rare reports of cyanocobalamin deficiency occurring with acid-suppressing therapy have been reported in the literature. This diagnosis should be considered if clinical symptoms consistent with cyanocobalamin deficiency are observed.

Interference with laboratory tests: Increased Chromogranin A (CgA) level may interfere with investigations for neuroendocrine tumours. If the patient(s) are due to have a test on Chromogranin A level, Zenpro treatment should be stopped for at least 5 days before CgA measurements to avoid this interference (See *Pharmacodynamics*). If CgA and gastrin levels have not returned to reference range after initial measurement, measurements should be repeated 14 days after cessation of proton pump inhibitor treatment.

Drug Interactions

Active substances with pH dependent absorption

The decreased intragastric acidity during treatment with omeprazole might increase or decrease the absorption of active substances with a gastric pH dependent absorption.

Nelfinavir, atazanavir

The plasma levels of nelfinavir and atazanavir are decreased in case of co-administration with omeprazole. Concomitant administration of omeprazole with nelfinavir is contraindicated. Co-administration of omeprazole (40 mg once daily) reduced mean nelfinavir exposure by ca. 40% and the mean exposure of the pharmacologically active metabolite M8 was reduced by ca. 75 - 90%. The interaction may also involve CYP2C19 inhibition. Concomitant administration of omeprazole with atazanavir is not recommended (see *Warnings and Precautions*). Concomitant administration of omeprazole (40 mg once daily) and atazanavir 300 mg/ritonavir 100 mg to healthy volunteers resulted in a 75% decrease of the atazanavir exposure. Increasing the atazanavir dose to 400 mg did not compensate for the impact of omeprazole on atazanavir exposure. The co-administration of omeprazole (20 mg once daily) with atazanavir 400 mg/ritonavir 100 mg to healthy volunteers resulted in a decrease of approximately 30% in the atazanavir exposureas compared to atazanavir 300 mg/ritonavir 100 mg once daily.

Digoxin

Concomitant treatment with omeprazole (20 mg daily) and digoxin in healthy subjects increased the bioavailability of digoxin by 10%. Digoxin toxicity has been rarely reported. However caution should be exercised when omeprazole is given at high doses in elderly patients. Therapeutic drug monitoring of digoxin should be then be reinforced.

Clopidogrel

Results from studies in healthy subjects have shown a pharmacokinetic (PK)/pharmacodynamic (PD) interaction between clopidogrel (300 mg loading dose/75 mg daily maintenance dose) and omeprazole (80 mg p.o. daily) resulting in a decreased exposure to the active metabolite of clopidogrel by an average of 46% and a decreased maximum inhibition of (ADP induced) platelet aggregation by an average of 16%. Inconsistent data on the clinical implications of a PK/PD interaction of omeprazole in terms of major cardiovascular events have been reported from both observational and clinical studies. As a precaution, concomitant use of omeprazole and clopidogrel should be discouraged (see *Warnings and Precautions*).

Other active substances

The absorption of posaconazole, erlotinib, ketoconazole and itraconazole is significantly reduced and thus clinical efficacy may be impaired. For posaconazole and erlotinib concomitant use should be avoided.

Active substances metabolised by CYP2C19

Omeprazole is a moderate inhibitor of CYP2C19, the major omeprazole metabolising enzyme. Thus, the metabolism of concomitant active substances also metabolised by CYP2C19, may be decreased and the systemic exposure to these substances increased. Examples of such drugs are R-warfarin and other vitamin K antagonists, clobazam, diazepam and phenytoin.

Cilostazol
Omeprazole, given in doses of 40 mg to healthy subjects in a cross-over study, increased Cmax and AUC for cilostazol by 18% and 26% respectively, and one of its active metabolites by 29% and 69% respectively.

Phenytoin
Monitoring phenytoin plasma concentration is recommended during the first two weeks after initiating omeprazole treatment and, if a phenytoin dose adjustment is made, monitoring and a further dose adjustment should occur upon ending omeprazole treatment.

Saquinavir
Concomitant administration of omeprazole with saquinavir/ritonavir resulted in increased plasma levels up to approximately 70% for saquinavir associated with good tolerability in HIV-infected patients.

Tacrolimus
Concomitant administration of omeprazole has been reported to increase the serum levels of tacrolimus. A reinforced monitoring of tacrolimus concentrations as well as renal function (creatinine clearance) should be performed, and dosage of tacrolimus adjusted if needed.

Methotrexate
When given together with proton-pump inhibitors, methotrexate levels have been reported to increase in some patients. In high-dose methotrexate administration a temporary withdrawal of omeprazole may need to be considered.

Inhibitors of CYP2C19 and/or CYP3A4
Since omeprazole is metabolised by CYP2C19 and CYP3A4, active substances known to inhibit CYP2C19 or CYP3A4 (such as clarithromycin and voriconazole) may lead to increased omeprazole serum levels by decreasing omeprazole's rate of metabolism. Concomitant voriconazole treatment resulted in more than doubling of the omeprazole exposure. As high doses of omeprazole have been well-tolerated adjustment of the omeprazole dose is not generally required. However, dose adjustment should be considered in patients with severe hepatic impairment and if long-term treatment is indicated.

Inducers of CYP2C19 and/or CYP3A4
Active substances known to induce CYP2C19 or CYP3A4 or both (such as rifampicin and St John's wort) may lead to decreased omeprazole serum levels by increasing omeprazole's rate of metabolism.

Pregnancy and Lactation

Pregnancy
Results from studies indicate no adverse effects of omeprazole on pregnancy or on the health of the foetus/newborn child. Omeprazole can be used during pregnancy.

Breast-feeding
Omeprazole is excreted in breast milk but is not likely to influence the child when therapeutic doses are used.

Side Effects / Adverse Reactions

Omeprazole is generally well tolerated.

Blood and lymphatic system disorders	
Rare:	Leukopenia, thrombocytopenia
Very rare:	Agranulocytosis, pancytopenia
Immune system disorders	
Rare:	Hypersensitivity reactions e.g. fever, angioedema and anaphylactic reaction/shock
Metabolism and nutrition disorders	
Rare:	Hyponatraemia
Not known:	Hypomagnesaemia; severe hypomagnesaemia may result in hypocalcaemia.
Psychiatric disorders	
Uncommon:	Insomnia
Rare:	Agitation, confusion, depression
Very rare:	Aggression, hallucinations
Nervous system disorders	
Common:	Headache
Uncommon:	Dizziness, paraesthesia, somnolence
Rare:	Taste disturbance
Eye disorders	
Rare:	Blurred vision
Ear and labyrinth disorders	
Uncommon:	Vertigo
Respiratory, thoracic and mediastinal disorders	
Rare:	Bronchospasm
Gastrointestinal disorders	
Common:	Abdominal pain, constipation, diarrhoea, flatulence, nausea/vomiting, fundic gland polyps (benign)
Rare:	Dry mouth, stomatitis, gastrointestinal candidiasis
Not known:	Microscopic colitis
Hepatobiliary disorders	
Uncommon:	Increased liver enzymes
Rare:	Hepatitis with or without jaundice
Very rare:	Hepatic failure, encephalopathy in patients with pre-existing liver disease
Skin and subcutaneous tissue disorders	
Uncommon:	Dermatitis, pruritus, rash, urticaria
Rare:	Alopecia, photosensitivity
Very rare:	Erythema multiforme, Stevens-Johnson syndrome, toxic epidermal necrolysis (TEN)
Not known:	Subacute cutaneous lupus erythematosus
Musculoskeletal and connective tissue disorders	
Uncommon:	Fracture of the hip, wrist or spine
Rare:	Arthralgia, myalgia
Very rare:	Muscular weakness
Renal and urinary disorders	
Rare:	Interstitial nephritis
Reproductive system and breast disorders	
Very rare:	Gynaecomastia
General disorders and administration site conditions	
Uncommon:	Malaise, peripheral oedema
Rare:	Increased sweating

Symptoms and Treatment for Overdosage

There is limited information available on the effects of overdoses of omeprazole in humans. In the literature, doses of up to 560 mg have been described, and occasional reports have been received when single oral doses have reached up to 2,400 mg omeprazole (120 times the usual recommended clinical dose). Nausea, vomiting, dizziness, abdominal pain, diarrhoea and headache have been reported. Also apathy, depression and confusion have been described in single cases. The symptoms described have been transient, and no serious outcome has been reported. The rate of elimination was unchanged (first order kinetics) with increased doses. Treatment, if needed, is symptomatic.

Shelf-life: The expiry date is indicated on the packaging.

Description: Size 2, brown-light brown hard gelatin capsules printed with "Zenpro 20".

Packing / Pack sizes: 2 x 7's, 4 x 7's and 5 x 7's in the aluminium-aluminium blister or strip. Not all packing(s) are available.

Storage: Store in a cool dry place below 25°C.

KEEP ALL MEDICINES OUT OF REACH OF CHILDREN/ JAUHI DARI KANAK-KANAK
For further information, please consult your pharmacist or physician.

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Manufacturer and Product Registration Holder:
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